

计算机网络练习题（选择专题）

II、Single Choice (1 point for each question)

1. Which of the following delays is variable (not constant) and depends on the current network traffic load?

- A) Processing delay
- B) Transmission delay
- C) Propagation delay
- D) Queuing delay

2. In the HTTP/1.1 protocol, what mechanism allows multiple objects to be sent over a single TCP connection?

- A) Non-persistent connection
- B) Persistent connection
- C) Web Caching
- D) Cookie exchange

3. When a TCP segment is lost, which mechanism initiates the Fast Retransmit?

- A) Time-out timer expiration
- B) Receipt of 3 duplicate ACKs
- C) Receipt of a FIN segment
- D) Window size dropping to zero

4. Suppose a DNS server does not know the IP address for a queried hostname. It asks another Root or TLD server on behalf of the client. This method is called:

- A) Iterative query
- B) Recursive query
- C) Caching query
- D) Reverse query

5. Which field in the IPv4 header is primarily responsible for handling fragmentation?

- A) Identification, Flags, Fragment Offset
- B) Header Checksum
- C) TOS (Type of Service)
- D) Source IP Address

6. In a "Store-and-Forward" packet-switched network, if a router has 3 packets of size L to send over a link with rate R , what is the total transmission delay for the 3 packets?

- A) L/R
- B) $2L/R$
- C) $3L/R$
- D) $L/3R$

7. Which routing algorithm suffers from the "Count-to-Infinity" problem?
- A) Link State (LS)
 - B) Distance Vector (DV)
 - C) OSPF
 - D) Dijkstra's Algorithm
8. What is the main purpose of the SEQ (Sequence Number) in TCP?
- A) To identify the application port.
 - B) To count the number of packets sent.
 - C) To order the bytes of data and detect duplicates/gaps.
 - D) To perform congestion control.
9. A switch in a LAN forwarding a frame with destination MAC FF:FF:FF:FF:FF:FF will:
- A) Drop the frame.
 - B) Forward it to the default gateway only.
 - C) Flood it to all interfaces except the incoming one.
 - D) Send it back to the sender.
10. In a subnet 192.168.1.0/26, how many usable host IP addresses are there?
- A) 254
 - B) 126
 - C) 62
 - D) 30
11. Why does the CSMA/CD protocol mandate a minimum frame size?
- A) To ensure the header overhead is not too large.
 - B) To allow the sender to detect a collision before it finishes transmitting the frame.
 - C) To make memory alignment easier in the NIC.
 - D) To prevent buffer underflow.
12. Which of the following is TRUE about Go-Back-N (GBN)?
- A) The receiver buffers out-of-order packets.
 - B) The sender maintains a timer for the oldest unacknowledged packet.
 - C) It is more efficient than Selective Repeat on high-error links.
 - D) It uses individual ACKs for every packet.
13. What is the value of the Protocol field in an IP header carrying a TCP segment?
- A) 6
 - B) 17
 - C) 1
 - D) 80

14. In Software Defined Networking (SDN), the "Control Plane" is typically:

- A) Distributed in every router.
- B) Centralized in a remote controller.
- C) Located in the physical link.
- D) Hard-coded in the switch firmware.

15. If a host has an IP address of 128.119.40.12 and a subnet mask of 255.255.255.0, which address belongs to the same subnet?

- A) 128.119.40.200
- B) 128.119.100.12
- C) 128.119.41.12
- D) 128.100.40.12

答案:

- 1.D 排队时延取决于流量, 不固定
- 2.B 持久连接
- 3.B 3 个冗余 ACK 触发快速重传
- 4.B 递归查询 - 服务器代替客户端去查
- 5.A 标识、标志、片偏移
- 6.C $3L/R$ 。虽然是存储转发, 但在单条链路上连续发 3 个包, 就是简单的序列化时间 $3 \times L/R$
- 7.B 距离向量 DV
- 8.C 字节排序和去重
- 9.C 广播/泛洪
- 10.C 主机位 6 位, $2^6 - 2 = 62$
- 11.B 边发边听, 必须在发送结束前检测到碰撞信号
- 12.B 基序号 Base 的定时器
- 13.A TCP 协议号为 6, UDP 为 17
- 14.B 集中式控制器
- 15.A 40.12 和 .40.200 的前 24 位相同

II. Single Choice (1 point for each question)

1. Which of the following is a key advantage of packet switching over circuit switching?
A) Guaranteed bandwidth for each call.
B) More efficient use of bandwidth through statistical multiplexing.
C) Lower packet loss rate.
D) Constant delay and jitter.

2. What is the primary function of a Cookie in the HTTP protocol?
A) To encrypt the data transfer.
B) To cache web pages on the client side to speed up loading.
C) To allow a stateless protocol (HTTP) to identify users and maintain session state.
D) To resolve the hostname to an IP address.

3. When a DNS server receives a query and returns the IP address of the authoritative name server for the requested domain (instead of the final IP), this is an example of:
A) Recursive resolution.
B) Iterative resolution.
C) DNS spoofing.
D) DNS caching.

4. Which of the following applications would prefer UDP over TCP?
A) Web browsing (HTTP).
B) File transfer (FTP).
C) Email (SMTP).
D) Real-time Voice over IP (VoIP).

5. In TCP Flow Control, the value of the Receive Window (rwnd) sent by the receiver tells the sender:
A) The size of the congestion window.
B) The maximum segment size (MSS).
C) The amount of free buffer space available at the receiver.
D) The sequence number of the next expected byte.

6. If a TCP sender detects packet loss via a Timeout, how does it adjust its parameters (in TCP Tahoe/Reno)?
A) Sets ssthresh to half of current cwnd, and sets cwnd to 1 MSS.
B) Sets ssthresh to half of current cwnd, and sets cwnd to ssthresh.
C) Decreases cwnd by 1 MSS.
D) Doubles the cwnd.

7. Which of the following is NOT a function of the Data Plane?

- A) Computing the shortest path from source to destination using OSPF.
- B) Checking the destination IP address of an incoming packet.
- C) Looking up the forwarding table.
- D) Moving the packet from an input port to the appropriate output port.

8. Which of the following is a valid Multicast IP address (Class D)?

- A) 192.168.1.1
- B) 127.0.0.1
- C) 224.0.0.5
- D) 10.0.0.5

9. Compared to IPv4, which field has been removed in the IPv6 header to speed up processing in routers?

- A) Source IP Address.
- B) Destination IP Address.
- C) Header Checksum.
- D) Hop Limit (TTL).

10. Why is Network Address Translation (NAT) controversial?

- A) It increases the number of available IPv4 addresses.
- B) It violates the end-to-end principle by modifying IP headers and port numbers.
- C) It makes the internal network completely visible to the outside world.
- D) It requires manual configuration on every host.

11. The MAC address is also known as:

- A) Logical Address.
- B) Port Address.
- C) Physical Address or LAN Address.
- D) URL.

12. What does an ARP Request packet contain?

- A) The MAC address of the destination, asking for its IP.
- B) The IP address of the destination, asking for its MAC.
- C) The domain name, asking for its IP.
- D) The routing table of the sender.

13. A switch is "transparent" to hosts. This means:

- A) Hosts must configure the switch's IP as their default gateway.
- B) Hosts communicate with the switch to reserve bandwidth.
- C) Hosts are unaware of the switch's presence; they send frames as if connected directly.
- D) The switch does not buffer frames.

14. Why does 802.11 (Wi-Fi) use CSMA/CA instead of CSMA/CD?
- A) Collisions never happen in wireless.
 - B) It is difficult for a radio to transmit and receive (listen) at the same time due to signal strength differences.
 - C) CA is faster than CD.
 - D) Wireless signals travel faster than light.
15. Consider a path with two links: Link 1 ($R_1=100\text{Mbps}$) and Link 2 ($R_2=10\text{Mbps}$). What is the maximum end-to-end throughput for a large file transfer?
- A) 110 Mbps.
 - B) 100 Mbps.
 - C) 55 Mbps.
 - D) 10 Mbps.

答案:

1. B: 分组交换通过统计多路复用 (Statistical Multiplexing) 允许按需共享链路, 因此带宽利用率比电路交换 (预留带宽) 更高。
2. C: HTTP 是无状态的, Cookie 用于在客户端和服务端之间维护状态 (如购物车、登录信息)。
3. B: 迭代解析 (Iterative)。服务器只返回下一级服务器的地址, 让查询者自己去问。如果是递归, 服务器会帮忙查到底。
4. D: 实时语音 (VoIP) 对延迟敏感, 但能容忍少量丢包, TCP 的重传机制会导致延迟抖动, 所以首选 UDP。
5. C: 接收窗口 (rwnd) 反映了接收方缓存区的剩余空间, 用于流量控制。
6. A: 在 TCP Tahoe/Reno 中, 发生超时 (Timeout) 时, 惩罚最重: ssthresh 设为当前 cwnd 的一半, cwnd 重置为 1 MSS, 重新进入慢启动。
7. A: 计算最短路径 (路由算法) 属于控制平面 (Control Plane) 的功能。数据平面只负责查表转发。
8. C: 224.0.0.0 到 239.255.255.255 是 D 类地址 (组播)。A 是 C 类单播私有, B 是回环, D 是 A 类单播私有。
9. C: IPv6 移除了 Header Checksum (首部校验和), 因为链路层和传输层已有校验, 移除可减少路由器处理时间 (每跳都要重新计算校验和太慢)。
10. B: NAT 破坏了 IP 的端到端原则, 路由器修改了报文头部的 IP 和端口, 导致某些特定应用 (如 P2P) 难以穿越。
11. C: 物理地址或 LAN 地址。
12. B: ARP 请求是广播发送: “我是 IP X, 谁是 IP Y? 请告诉我你的 MAC。”
13. C: 透明性意味着主机不需要知道交换机的存在, 也不需要配置交换机地址, 即插即用。
14. B: 无线电通信中, 发送信号强度远大于接收信号, 导致设备在发送时无法检测到远端的微弱碰撞信号 (隐蔽站问题也是原因之一, 但无法边发边听是物理限制)。
15. D: 吞吐量受限于瓶颈链路 (最慢的一段)。 $\min(100, 10) = 10\text{Mbps}$ 。

II、Single Choice (1 point for each question)

1. In pipelining protocols, the selective repeat approach requires ().
- A、 receiver acks individual packets
- B、 sender maintains timer for cumulative unacked packets
- C、 receiver only sends cumulative acks
- D、 both B and C responses are correct
2. Which application's NOT using TCP? ()
- A、 SMTP B、 HTTP C、 DNS D、 All of them
3. What are the advantages of Distance Vector Routing over Link State Routing? ()
- A、 Changes faster, especially when links go down.
- B、 Requires less computation in each router.
- C、 The computed routes are of better quality.
- D、 Requires more information to be exchanged by the routers.
4. The header field based on which the UDP protocol implements demultiplexing is ().
- A、 source port number B、 destination port number C、 length D、 checksum
5. Table 1 shows the routing table of a router. If a router receives an IP packet whose destination address is 169.96.40.5, the interface that forwards the IP packet is ().
- A、 S1 B、 S2 C、 S3 D、 S4

Table 1

Destination Network	Next Hop	Interface
169.96.40.0/23	176.1.1.1	S1
169.96.40.0/25	176.2.2.2	S2
169.96.40.0/27	176.3.3.3	S3
0.0.0.0/0	176.4.4.4	S4

6. Which DNS record type is used to associate an IP address with a domain name?
- A) A record

- B) MX record
- C) CNAME record
- D) NS record

7. What does an HTTP 304 response code indicate?

- A) The requested resource has been permanently moved to a new location.
- B) The server encountered an error while processing the request.
- C) The requested resource has not been modified since the last request.
- D) The server is temporarily unavailable or overloaded.

8. Which type of applications or services are commonly associated with UDP usage?

- A) File transfer and email services
- B) Web browsing and HTTP traffic
- C) Video streaming and online gaming
- D) Remote shell access and Telnet

9. What happens when the receiver's buffer is full during TCP flow control?

- A) The receiver discards incoming data packets.
- B) The sender reduces its data transmission rate.
- C) The receiver sends a congestion notification to the sender.
- D) The sender increases the size of the sliding window.

10. The characteristics of the UDP protocol include:

- A) Detection of bit errors
- B) Congestion control
- C) Flow control
- D) Bandwidth guarantee

答案:

1. A 在选择重传 (SR) 协议中, 接收方对每个正确接收的分组单独发送 ACK, 而不是累积 ACK。
2. C DNS 主要使用 UDP (端口 53), 虽然在区域传送时用 TCP, 但常规查询用 UDP。SMTP 和 HTTP 完全依赖 TCP。
3. B 距离向量 (DV) 算法相较于链路状态 (LS) 算法, 每个节点仅需与邻居交换信息, 且计算相对简单 (LS 需要 Dijkstra 算法)。
4. B UDP 解复用 (Demultiplexing) 主要依据目的端口号和目的 IP 地址 (如果是连接无关, 主要看目的端口)。TCP 才需要四元组。
5. C
S1 (169.96.40.0/23): 范围 .40.0 - .41.255。匹配。
S2 (169.96.40.0/25): 范围 .40.0 - .40.127。匹配。最长前缀匹配。
S3 (169.96.40.0/27): 范围 .40.0 - .40.31。5 在范围内。匹配。
比较前缀长度: /23, /25, /27。/27 最长, 所以选 S3。
6. A A 记录将域名映射为 IPv4 地址。
7. C 304 Not Modified (未修改, 使用缓存)。
8. C 视频流和游戏 (容忍丢失, 对延迟敏感)。
9. B 流量控制中, 接收窗口满, 发送方将停止发送数据 (实际上是发送窗口变 0, 导致发送速率降为 0)。选项 B “降低速率” 最接近。
10. A UDP 只提供校验和 (Bit error detection)。无流控、无拥塞控制、无带宽保证。